



REFERENCE ARCHITECTURE

Programmable transport for federal data centers, cloud on-ramps, and edge

MaiaEdge for Federal Data Center Operators.

A reference architecture for federal data center interconnect,
cloud on-ramps, and edge deployments.

Outer transport on Intel IPU silicon. Layer 2 transparent.
Software-defined diversity across fiber, LEO, and commercial internet.

Private paths. Any network. Instantly.

Carrier infrastructure for federated private networking

Executive Summary

The federal data center landscape today is distributed across agencies, multi-cloud, mission-critical, and operated under tight modernization mandates. Workloads move between owned data centers, hyperscaler GovCloud regions, and field-deployed sites. The architecture above the transport layer has matured considerably. The transport layer itself has not.

Cross-connects between federal data centers still run on 60-to-90-day timelines. Cloud on-ramps to GovCloud, Azure Government, and GCP Government are bespoke per-program builds. Field-deployed kits wait on fiber installs that take longer than the missions they support. Each new diverse path is its own L3 integration project. The operator's engineers spend their time on routing-protocol toil instead of customer outcomes.

MaiaEdge is software-defined carrier infrastructure built on Intel Infrastructure Processing Unit (IPU) silicon. It collapses the transport layer into a programmable fabric: appliances at every site execute path instructions from a customer-managed control plane, encrypt at line rate, and ride a blended underlay of fiber, LEO satellite, and commercial internet. The architecture sits cleanly beneath inline Type-1 encryption when a program requires it, and delivers equivalent operational benefits in programs where Type-1 is not in scope.

This paper describes where the architecture fits in operator programs, how it is composed, and the posture under which it deploys.

02 WHERE IT GATES

Where Transport Friction Shows Up in Operator Work

Transport is the layer that bottlenecks federal modernization programs in four predictable places.

Federal data center interconnect. Operators run estates of data centers spanning DoD, civil agency, and intelligence community customers. Cross-connects between those data centers are still provisioned manually, with weeks of LOA paperwork, BGP work, and meet-me-room coordination per circuit. Tenants who want self-service capacity wait on the operator's queue.

Cloud on-ramps. Federal customers consume capacity from AWS GovCloud, Azure Government, GCP Government, and Oracle Government in parallel. Each on-ramp is a per-program engineering project. The operator's brand sits in front; the operator's engineers carry the BGP, the failover policy, and the SLA accountability.

Field-deployed and globally-shipped systems. Mission-critical kits get racked, packed, and shipped to federal sites worldwide. The transport plan typically waits until the kit lands. Programs lose weeks while fiber, MPLS, or carrier circuits are arranged at the destination.

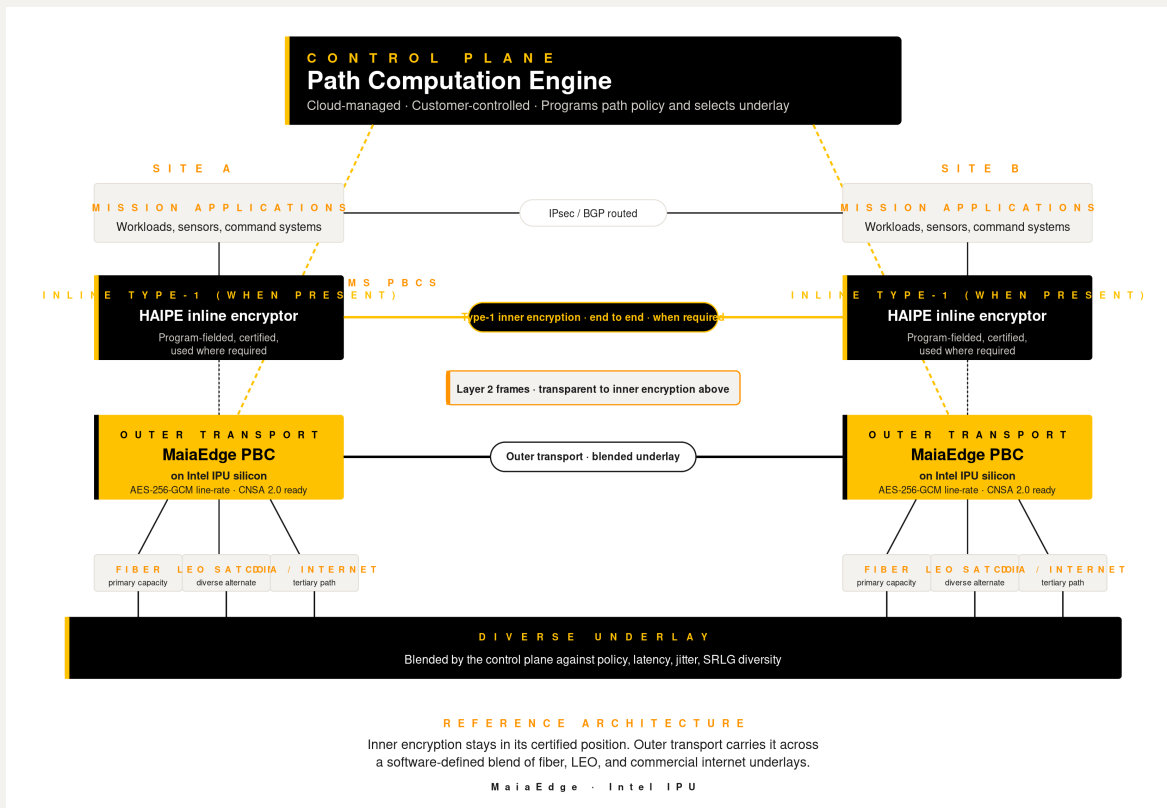
Inter-agency and cross-domain connectivity. Programs that span agencies have to coordinate transport across multiple operators, multiple certification regimes, and multiple control planes. The underlay always exists; the unified visibility and policy enforcement above it usually does not.

Each of these is a place where the application architecture is ready, the customer is ready, and the program timeline is gated by transport. The reference architecture in this paper moves transport off the critical path.

03 ARCHITECTURE

The Reference Architecture

The architecture is composed of an outer transport function on Intel IPU silicon at each network boundary, a path computation engine running in operator-managed cloud, and a Layer 2 transparency property that lets anything above the outer transport, including IPsec tunnels, BGP-routed networks, and inline Type-1 encryption when a program requires it, behave as if it were on a single end-to-end Ethernet path.



Outer transport on the IPU

A 1RU appliance at each network boundary. Dual 100 Gbps interfaces. Stateless forwarding. Line-rate AES-256-GCM encryption. Sub-2-microsecond latency overhead. The hardware refresh in Q1 2027 takes the line rate to 200 Gbps. The appliance executes path instructions; it does not compute them.

Path computation engine in operator-managed cloud

A cloud-native engine that maintains a global view of the topology, computes deterministic end-to-end paths against policy, SLA, and SRLG constraints, programs the appliances over a secure channel, and collects hop-by-hop telemetry continuously. The control plane runs in the operator's own cloud environment. No data-plane traffic transits the control plane.

Layer 2 transparency, with or without inline Type-1

The outer transport carries Layer 2 frames end to end. Programs that field inline Type-1 encryption (HAiPE-class) keep that stack exactly where it sits today, with no certification impact and no new component in the certified path. Programs that do not require Type-1 inherit the same architectural properties: hardware-accelerated encryption at line rate, deterministic transport behavior, and a blended underlay that the operator manages programmatically. The architecture makes inline Type-1 easy to live with where it is required, and earns its keep in programs where it is not.

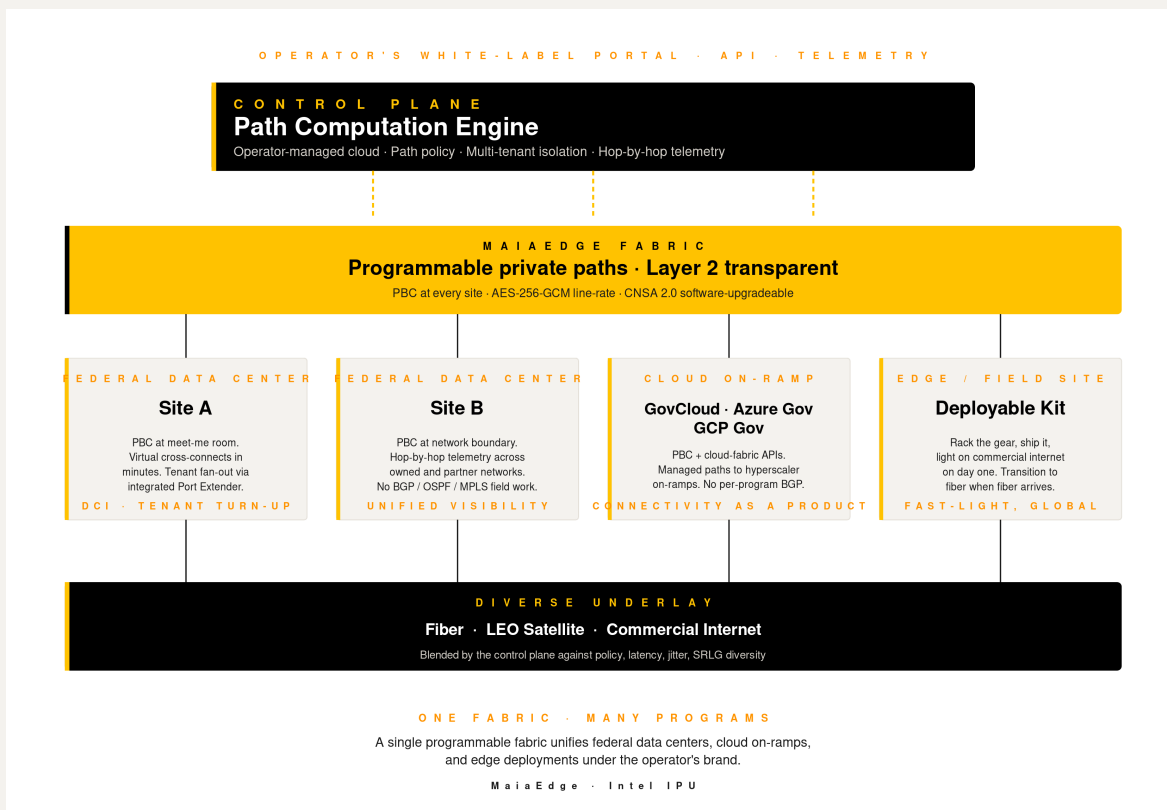
CNSA 2.0 readiness

The Commercial National Security Algorithm Suite 2.0 mandate takes effect January 1, 2027. The reference architecture is software-upgradeable to CNSA 2.0 algorithms in the data plane. Programs deploying it today inherit a path to conformance ahead of the mandate.

04 USECASES

What This Looks Like in Operator Programs

The reference architecture lands in four places in the operator's motion. Each places one appliance at a site boundary, claims it under the operator's control plane, and unifies it with everything else the operator runs.



Federal data center interconnect. Appliances at the meet-me room of each data center turn the operator's footprint into a programmable fabric. Virtual cross-connects activate in minutes. Tenant fan-out at high port density runs on the integrated Port Extender. Hop-by-hop telemetry covers owned and partner networks alike. The customer self-serves through the operator's branded portal.

Cloud on-ramps. A single appliance at the cloud-fabric handoff, paired with API integrations to Equinix Fabric, Megaport, AWS Direct Connect, Azure ExpressRoute, and equivalents, turns cloud connectivity into a managed product the operator delivers under its own brand. No per-program BGP integration. No bespoke pipeline per customer. The operator owns the customer relationship and the margin.

Field-deployed and globally-shipped systems. Operators rack the kit, pack it, and ship it to the destination on the program's schedule. The appliance lights on commercial internet on day one, adds LEO satellite as a diverse alternate, and transitions to fiber when fiber arrives, all without re-architecture. Programs that used to wait three months for fiber start operating in days.

Inter-agency and cross-domain connectivity. When a program spans multiple agencies, the same fabric extends across the boundary cleanly. The control plane enforces policy, jurisdictional routing, and SLA constraints programmatically. Audit trails are hop-by-hop and exportable. The operator's engineers do not have to re-engineer the underlay each time the program scope expands.

In all four cases, the operator's engineering effort moves from routing-protocol work to customer outcomes. The hardware ownership stays with MaiaEdge under a subscription model, fitting the operator's OpEx posture.

05 POSTURE

Posture and Operating Model

The reference architecture is designed to meet operator programs where they are, rather than impose a single deployment or commercial model.

Available today. The outer transport appliance is operational and shipping. The control plane runs on AWS today; the same image deploys into AWS GovCloud. 100% US-developed software, 100% US-designed hardware integrated in the United States on Intel IPU silicon, 100% US-based engineering team. Customer-controlled deployment posture: the

implementer does not require visibility into operator data planes, traffic, or operational telemetry.

Adaptable per program requirements. FIPS 140-3 validation through third-party accreditation is achievable; MaiaEdge undertakes that work as a program requires it. IL5 deployment is open and would be pursued alongside program-specific accreditation. The control plane can be adapted to operator-managed environments outside AWS through targeted engineering scoped to the program.

Commercial model. Annual subscription, hardware title held by MaiaEdge, software updates and support included. This fits the operator service model where capital constraints and OpEx accounting are operationally important.

06 IMPLEMENTER

About MaiaEdge

MaiaEdge is carrier infrastructure for federated private networking. The founding team built two prior carrier-infrastructure companies: Acme Packet, the Session Border Controller used by 90% of global carriers, sold to Oracle for \$2.1 billion; and 128 Technology, a software-defined wide-area transport platform fielded across all branches of the United States Department of Defense, acquired by Juniper. Two exits, \$2.5 billion+ combined, in this exact category of infrastructure.

100% US-developed software. 100% US-designed hardware integrated in the United States on Intel IPU silicon. 100% US-based engineering team. Headquartered in Boston, Massachusetts.

The company welcomes a working session with operator engineering and innovation teams to evaluate the reference architecture against the programs that matter most to your customers.

07 CONTACT

For Technical Conversations

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